

Overview:

Course: Azure CloudLabs [AZ-104, AZ-204, AZ-305]

This document outlines the resources allowed and denied for hands-on lab environments. The lab restrictions are aligned to the lab exercises, so by keeping in mind the allowed SKUs/Resource you will be able to complete the lab exercises without any kind of restrictions.

Allowed SKUs & Resources

1. Azure Windows Virtual Machine

a. Allowed:

- i. VMs with 1 & 2 vCPUs - Standard_B1s, Standard_B2s, Standard_B2ms, Standard_DS1_v2, Standard_DS2_v2 & Standard_D2s_v3
- ii. Images –
 1. Windows 2019 & 2016 data center gen2
 2. Windows 2019-Datacenter
 3. Windows 10 Pro
 4. Windows 2016-Datacenter
 5. Windows 2016-Datacenter-Server-Core,
 6. Windows 2016-Datacenter-Server-Core-smalldisk

Note : Any version of Windows image can be deployed for the VMs

b. Denied:

- i. All other sizes except for the ones mentioned above are denied
- ii. All Windows images are denied except for the ones mentioned in the Allowed section

c. Recommendation

- i. Please use the sizes Standard_B1s, Standard_B2s, Standard_B2ms, Standard_DS1_v2, Standard_DS2_v2 & Standard_D2s_v3s

- ii. Please select image- Windows Server 2019 Datacenter - x64 Gen2, Windows Server 2016 Datacenter- x64 Gen2

2. Azure Linux Virtual Machine

- a. Allowed:
 - i. VMs with 1 & 2 vCPUs - Standard_B1s, Standard_B2s, Standard_B2ms, Standard_DS1_v2, Standard_DS2_v2 & Standard_D2s_v3
 - ii. Images –
 - 1. Ubuntu 20.04 LTS, 18.04 LTS & Minimal 18.04 LTS
- b. Denied:
 - i. All other sizes except for the ones mentioned above are denied
 - ii. All Linux images are denied except for the ones mentioned in the recommendation.
- c. Recommendation
 - i. Please use the sizes Standard_B1s, Standard_B2s, Standard_B2ms, Standard_DS1_v2, Standard_DS2_v2 & Standard_D2s_v3
 - ii. Please select image- Ubuntu Server 20.04 LTS – x64 Gen2

3. Disks

- a. Allowed:
 - i. Standard HDD
- b. Denied:
 - i. All others.
- c. Recommendation:
 - i. Please select the OS disk type as Standard HDD while creating a VM as it is cost effective.

4. Azure Virtual Machine Scale Set

- a. Allowed sku:
 - i. VMs with 1 & 2 vCPUs - Standard_B1s, Standard_B2s, Standard_B2ms, Standard_DS1_v2, Standard_DS2_v2 & Standard_D2s_v3
 - ii. Capacity 1 or 2 VM instances.
- b. Denied:

- i. All other sizes except for the ones mentioned above are denied
- c. Recommendation
 - i. Please use the sizes Standard_B1s, Standard_B2s, Standard_B2ms, Standard_DS1_v2, Standard_DS2_v2 & Standard_D2s_v3
 - ii. Please keep the Instances count as 1 only.
 - iii. Please deploy VM Disk with Standard HDD only.
 - iv. For Windows Images please select image- Windows Server 2019 Datacenter - x64 Gen2, Windows Server 2016 Datacenter- x64 Gen2
 - v. For Ubuntu Images please select image- Ubuntu Server 20.04 LTS – x64 Gen2

5. Azure Virtual Network

- a. Allowed:
 - i. All configuration/properties allowed
- b. Denied:
 - i. DDOS creation
 - ii. Express Route Circuits creation
 - iii. Local Network Gateways creation
 - iv. Network Watchers creation

6. Public IP

- a. Recommendation
 - i. Please deploy public ip with Basic sku only.

7. ARM:

- a. Allowed:
 - i. Resources mentioned in the learner guide with specific sizes.
- b. Denied:
 - i. Restricted deployment of all the other resources not mentioned in this guide.

8. VNet Peering

- a. Recommendation:
 - i. Please do the peering between Vnets in the same region.

9. Diagnostic Settings

- a. Recommendation
 - i. Enable with managed storage account instead of custom storage account

10. Azure Monitor: Action Group, Azure Alerts, Monitor Metrics, Log Analytics

- a. Allowed
 - i. All configurations
- b. Denied
 - i. No Restrictions

11. Generate Report on Spent

- a. Allowed
 - i. Allowed permission to cost analysis

12. Azure Storage Account

- a. Allowed:
 - i. All configuration/properties allowed
 - ii. Network Access also allowed.
- b. Denied
 - i. No Restrictions
- c. Recommendation
 - i. Do not enable Defender for Storage account as that is not required as part of lab.
 - ii. Please use redundancy: LRS for cost optimization.

13. Azure AD- Add a custom domain: It is not possible to add custom domain.

14. Azure AD Join

- a. Allowed Image
 - i. Windows 10 pro
- b. Recommendation:
 - i. Use the windows server created earlier as domain controller.

15. Azure Site Recovery

- a. Recommendation:
 - i. Only backup VM

16. VPN Gateway

- a. Allowed sku
 - i. Sku tier- Basic
 - ii. Gateway type- Vpn
 - iii. Vpn Type- RouteBased
- b. Denied
 - i. All others.
- c. Recommendation
 - i. Please deploy the Basic Generation 1 route based VPN Gateway

17. Azure Load Balancer

- a. Allowed sku:
 - i. Basic
 - ii. Standard
- b. Denied:
 - i. All others.
- c. Recommended:
 - i. We are allowing Standard load balancer only for Azure Kubernetes service deployment to be succeeded since the managed RG uses the above-mentioned load balancer SKU.
 - ii. Please select Basic sku while deploying load balancer as it is free whereas Standard Load Balancer is costly

18. Azure Application Gateway

- a. Allowed sku:
 - i. Tier- Standard_v2
 - ii. Instance count- 1
 - iii. Sku size- Standard_v2
- b. Deny sku:
 - i. All others

- c. Recommendation
 - i. It is very costly resource please use carefully.

19. Traffic Manager

- a. Allowed
 - i. All configuration/properties allowed.
- b. Denied
 - i. No Restrictions

20. Azure Front Door

- a. Allowed sku:
 - i. Standard
- b. Denied sku:
 - i. Premium

21. Azure Logic Apps

- a. Allowed Plan:
 - i. Consumption
- b. Denied plan:
 - i. Standard

22. Azure Functions

- a. Allowed Plan:
 - i. Consumption
- b. Denied Plan:
 - i. All others.

23. Azure Event Grid: First 100,000 operations are free

24. Azure Service Bus

- a. Allowed namespace sku:
 - i. Basic
- b. Denied namespace sku:ap
 - i. Standard
 - ii. Premium

- c. Recommendation:
 - i. Please deploy queue with 1GB size

25. Azure Policy Creation: **Not Allowed**. Any activities related to changing Azure role/rights/restrictions are not allowed, as this possesses a risk of students removing the restrictions in-place

26. Azure Policy Assignment: **Not Allowed**. Any activities related to changing Azure role/rights/restrictions are not allowed, as this possesses a risk of students removing the restrictions in-place

27. Azure-AD User Creation

- a. Allowed:
 - i. Allowed permission to create user.
- b. Denied:
 - i. No restrictions.

28. Azure-AD Guest User Creation

- a. Allowed:
 - i. Allowed permission to create guest user.
- b. Denied:
 - i. No restrictions.

29. Azure-AD Group Creation

- a. Allowed:
 - i. Allowed permission to create group.
- b. Denied:
 - i. No restrictions.

30. Azure-AD Self Service Password Reset

- a. Allowed:
 - i. Allowed permission to enable SSPR.
- b. Denied:
 - i. No restrictions.

31. Azure Role Based Access Control: Not Allowed. Any activities related to changing Azure role/rights/restrictions are not allowed, as this possesses a risk of students removing the restrictions in-place

32. Azure App service

- a. Allowed:
 - i. Plan – Free, Basic
- b. Denied:
 - i. Plan - All others
- c. Recommendation:
 - i. Please use Free wherever possible.

33. Web App for container

- a. Allowed:
 - i. Plan - Free & Basic
- b. Denied:
 - i. Plan - All others
- c. Recommendation
 - i. Please select Linux OS for docker container. This will allow you to select free plan for app service.

34. Azure Container Instance

- a. Allowed:
 - i. Standard Sku
 - ii. 1, 1.5 GB Memory
 - iii. 1 Core vCPU
- b. Denied:
 - i. All others.

35. Azure Container Registry

- a. Allowed:
 - i. Sku - Basic
- b. Denied:
 - i. Sku - All others.

36.Azure Kubernetes service

- a. Allowed:
 - i. Pricing tier - Free
 - ii. Sku - Standard_B2s, Standard_B2ms, Standard_DS2_v2 & Standard_D2s_v3
- b. Denied:
 - i. Pricing tier - All others.
- c. Recommendation
 - i. Please deploy Linux AKS cluster for cost optimization.
 - ii. Please do not add more node pools.
 - iii. Please set the Scaling method as Manual.
 - iv. Please keep the node count as 1.

37.Azure Key Vault

- a. Allowed:
 - i. Sku - Standard
- b. Denied:
 - i. Sku - All others

38.Azure Event Hub

- a. Allowed:
 - i. Sku - Basic
 - ii. Capacity - 1
- b. Denied:
 - i. Sku - All others.

39.Azure SQL database

- a. Allowed:
 - i. Basic
 - ii. Local Redundancy.
- b. Denied:
 - i. All others.
- c. Recommendation
 - i. Please deploy database with LRS for Backup redundancy.

40. Azure Firewall

- a. Allowed:
 - i. Sku - Basic
- b. Denied:
 - i. Sku – Standard & Premium

41. Azure Notification Hub

- a. Allowed:
 - i. Pricing tier - Free
- b. Denied:
 - i. Pricing tier – Basic & Standard

42. Azure Cache for Redis

- a. Allowed:
 - i. Cache type - Basic c0
- b. Denied:
 - i. Cache type - All others.

43. Azure cosmos DB for NoSQL

- a. Allowed:
 - i. Capacity mode - Serverless
- b. Denied:
 - i. Capacity mode - Provisioned throughput
- c. Recommendation
 - i. Please deploy database with LRS for Backup redundancy.

44. Resource Group

- a. Allowed:
 - i. We have allowed students to create resource group

45. Azure AD PIM & Azure AD Identity Protection: Not Allowed. Any activities related to changing Azure role/rights/restrictions are not allowed, as this possesses a risk of students removing the restrictions in-place.

